

1 Gain law

(bars denote normalized quantities: lengths by R , $\bar{a} = a/a_o$, $\bar{f} = a_o f$; $\bar{w} > \bar{w}_s$, $b > 0$, $p > 0$; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}(\bar{w} - \bar{w}_s, b, p) = 1 - \left(1 + \frac{\bar{w} - \bar{w}_s}{b}\right)^{-p} \quad (\bar{w}_s - 1 : \text{unknown surface position})$$

$$\bar{a}(\bar{w} - \bar{w}_s, b_w, p_w) = 1 - \left(1 + \frac{\bar{w} - \bar{w}_s}{b_w}\right)^{-p_w}$$

$$\bar{a}(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = 1 - \left(1 + \frac{\bar{w}_d - \hat{w}_s}{\hat{b}_w}\right)^{-\hat{p}_w}$$

$$a_w = a_o \bar{a}(\bar{w} - \bar{w}_s, b_w, p_w), \quad a_o = \frac{\Delta t}{\gamma_N R} \quad (a_o \text{ fixed, not a state})$$

$$p = 1 : \quad \bar{a} = \frac{\bar{w} - \bar{w}_s}{(\bar{w} - \bar{w}_s) + b}, \quad \frac{d\bar{a}}{d\bar{w}} = \frac{b}{((\bar{w} - \bar{w}_s) + b)^2}$$

Parallel-axis law (2026-08-02, Stage 1a. The x/y truth $c_\parallel$ leaves contact with Goldman log behavior — no family member reproduces it, and the best representation places its origin INSIDE the wall: the parallel axes cannot report the physical wall. Constants derived at init from the planned envelope (offline truth-curve fit, the envelope-prior mechanism); the x/y filters lock $(b, p, \bar{w}_s)$ and consume the SHARED wall estimate one-way from the z filter; rationale, widths and health record in ref)

$$\bar{a}_\parallel = 1 - \left(1 + \frac{\bar{w} - (\hat{w}_{s,z} + \Delta_\parallel)}{b_\parallel}\right)^{-p_\parallel}, \quad \Delta_\parallel := \bar{w}_{s0,\parallel} - 1$$

$$(b_\parallel, p_\parallel, \bar{w}_{s0,\parallel}) = \arg \min \sup_{\text{envelope}} |\bar{a}_\parallel - 1/c_\parallel| = (0.5217, 0.9770, 0.5918) \quad (\sup 0.038\%)$$

$$P_{\bar{a}\bar{a},\parallel}[0] = (\text{shape floor}_\parallel)^2,$$

$$\text{floor}_\parallel = \sqrt{\sup_{fit}^2 + (\sup \bar{a}' \delta_{w_{s0}})^2 + (\sup |\frac{g}{b} \bar{a}'| \delta_b)^2 + (\sup |(1-\bar{a}) \ln(1+\frac{g}{b})| \delta_p)^2} = 0.0052$$

(boundary widths $\delta_b, \delta_p, \delta_{w_{s0}} = 0.0104, 0.0061, 0.0147$ — floor ± 0.1 / ceiling refits; the z law and the z filter are untouched)

2 Gain rate and parameter Jacobians

$$\bar{a}'(\bar{w} - \bar{w}_s, b, p) := \frac{d\bar{a}}{d\bar{w}} = \frac{p}{b} \left(1 + \frac{\bar{w} - \bar{w}_s}{b}\right)^{-p-1}$$

$$\bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) = \frac{\hat{p}_w}{\hat{b}_w} \left(1 + \frac{\bar{w}_d - \hat{w}_s}{\hat{b}_w}\right)^{-\hat{p}_w-1}$$

$$\frac{\partial \bar{a}}{\partial \bar{w}_s} = -\bar{a}' \quad (\text{exact, any } b, p)$$

$$\begin{aligned} & \bar{a}'(\bar{w}_d - \bar{w}_s, b_w, p_w) - \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) \\ &= J_b(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) e_{b_w} + J_p(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) e_{p_w} + J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) e_{\bar{w}_s} \end{aligned}$$

$$\begin{aligned} J_b(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) &= \left[-\frac{1}{\hat{b}_w} + \frac{(\hat{p}_w + 1)(\bar{w}_d - \hat{w}_s)}{\hat{b}_w((\bar{w}_d - \hat{w}_s) + \hat{b}_w)} \right] \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) \\ J_p(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) &= \left[\frac{1}{\hat{p}_w} - \ln\left(1 + \frac{\bar{w}_d - \hat{w}_s}{\hat{b}_w}\right) \right] \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) \\ J_{w_s}(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) &= \left[\frac{\hat{p}_w + 1}{(\bar{w}_d - \hat{w}_s) + \hat{b}_w} \right] \bar{a}'(\bar{w}_d - \hat{w}_s, \hat{b}_w, \hat{p}_w) \end{aligned}$$

3 True dynamics

$$(\delta \bar{w}_j[k] := \delta \bar{w}[k - d + j - 1], \quad j = 1, \dots, d+1, \quad d = 2; \quad \delta \bar{w}_D \equiv 0 \text{ hereafter})$$

$$\begin{aligned} \bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\ \bar{a}_w[k+1] &= \bar{a}_w(\bar{w}[k+1]) = \bar{a}(\bar{w}_d[k+1] - \bar{w}_s - \delta \bar{w}_3[k+1], b_w, p_w) \\ \bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta \bar{w}_d[k] \end{aligned}$$

$$\begin{aligned} \delta \bar{w}_1[k+1] &= \delta \bar{w}_2[k] \\ \delta \bar{w}_2[k+1] &= \delta \bar{w}_3[k] \\ \delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \quad (e_{\bar{a}_w} = \bar{a}_w - \hat{a}_w) \\ b_w[k+1] &= b_w[k] \\ p_w[k+1] &= p_w[k] \\ \bar{w}_s[k+1] &= \bar{w}_s[k] \end{aligned}$$

$$\begin{aligned} F_{dw}[k] &:= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\ \varepsilon_w[k] &:= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] - (1 - \lambda_c) n_w[k-d] \end{aligned}$$

(sign and lag of the n_w feedthrough corrected 2026-08-01 by log forensics — regression tap -0.296 at $k-d$; the former $+(1 - \lambda_c)n_w[k]$ leaves the variance unchanged, see ref)

4 Process model

$$\bar{w}[k+1] - \bar{w}[k] = \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

$$\bar{a}'[k] = \bar{a}'(\bar{w}_d[k] - \bar{w}_s[k], b_w[k], p_w[k])$$

$$\mathbf{x}[k] = [\delta \bar{w}_1, \delta \bar{w}_2, \delta \bar{w}_3, \bar{a}_w, b_w, p_w, \bar{w}_s]^\top [k] \quad (d = 2)$$

$$\delta \bar{w}_1[k+1] = \delta \bar{w}_2[k]$$

$$\delta \bar{w}_2[k+1] = \delta \bar{w}_3[k]$$

$$\delta \bar{w}_3[k+1] = \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\bar{a}_w[k+1] = \bar{a}_w[k] + \bar{a}'[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \right]$$

$$b_w[k+1] = b_w[k]$$

$$p_w[k+1] = p_w[k]$$

$$\bar{w}_s[k+1] = \bar{w}_s[k]$$

5 Estimator

$$\delta \hat{w}_1[k+1] = \delta \hat{w}_2[k]$$

$$\delta \hat{w}_2[k+1] = \delta \hat{w}_3[k]$$

$$\delta \hat{w}_3[k+1] = \lambda_c \delta \hat{w}_3[k]$$

$$\hat{a}_w[k+1] = \hat{a}_w[k] + \hat{a}'[k] \left[\Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \hat{w}_3[k] \right]$$

$$\hat{b}_w[k+1] = \hat{b}_w[k]$$

$$\hat{p}_w[k+1] = \hat{p}_w[k]$$

$$\hat{w}_s[k+1] = \hat{w}_s[k]$$

$$\hat{a}'[k] = \hat{a}'(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k])$$

6 Error dynamics

(True – Estimator; coefficients at the estimate)

$$e_{\delta\bar{w}_j} = \delta\bar{w}_j - \delta\hat{w}_j \quad (j = 1, 2, 3), \quad \mathbf{e} = [e_{\delta\bar{w}_1}, e_{\delta\bar{w}_2}, e_{\delta\bar{w}_3}, e_{\bar{a}_w}, e_{b_w}, e_{p_w}, e_{\bar{w}_s}]^\top$$

$$e_{\delta\bar{w}_1}[k+1] = e_{\delta\bar{w}_2}[k]$$

$$e_{\delta\bar{w}_2}[k+1] = e_{\delta\bar{w}_3}[k]$$

$$e_{\delta\bar{w}_3}[k+1] = \lambda_c e_{\delta\bar{w}_3}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k]$$

$$\begin{aligned} e_{\bar{a}_w}[k+1] = & [1 + \hat{a}'[k] F_{dw}[k]] e_{\bar{a}_w}[k] + J_{b_w}[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] e_{b_w}[k] \\ & + J_{p_w}[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] e_{p_w}[k] + J_{w_s}[k] [\Delta\bar{w}_d[k] + (1 - \lambda_c) \delta\hat{w}_3[k]] e_{\bar{w}_s}[k] \\ & + (1 - \lambda_c) \hat{a}'[k] e_{\delta\bar{w}_3}[k] + \hat{a}'[k] \varepsilon_w[k] \end{aligned}$$

$$e_{b_w}[k+1] = e_{b_w}[k]$$

$$e_{p_w}[k+1] = e_{p_w}[k]$$

$$e_{\bar{w}_s}[k+1] = e_{\bar{w}_s}[k]$$

$$J_{b_w}[k] = J_b(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k])$$

$$J_{p_w}[k] = J_p(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k])$$

$$J_{w_s}[k] = J_{w_s}(\bar{w}_d[k] - \hat{w}_s[k], \hat{b}_w[k], \hat{p}_w[k])$$

$$\mathbf{e}[k+1] = \mathbf{F}_e[k] \mathbf{e}[k] + \mathbf{q}[k]$$

$$\mathbf{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & \lambda_c & -F_{dw} & 0 & 0 & 0 \\ 0 & 0 & (1-\lambda_c)\hat{a}' & 1+\hat{a}'F_{dw} & J_{b_w}[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{w}_3] & J_{p_w}[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{w}_3] & J_{w_s}[\Delta\bar{w}_d+(1-\lambda_c)\delta\hat{w}_3] \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathbf{q}[k] = [0, 0, -\varepsilon_w[k], \hat{a}'[k] \varepsilon_w[k], 0, 0, 0]^\top \quad (q_4 = -\hat{a}'[k] q_3)$$

7 Measurement and innovation

Measurement (d -step delay; $\bar{a}_{wm} = a_{wm}/a_o$; y_2 raw readout — AR(1) whitening enters with R_2 , S8)

$$\nabla_d \bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]$$

$$y_1[k] = \delta\bar{w}_m[k] = \delta\bar{w}_1[k] + n_w[k]$$

$$y_2[k] = \bar{a}_{wm}[k] = \bar{a}_w[k-d] + n_a[k]$$

Delay back-off

$$\bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}'[k] \nabla_d \bar{w}_d[k] \quad \implies \quad y_2[k] = \bar{a}_w[k] - \bar{a}'[k] \nabla_d \bar{w}_d[k] + n_a[k]$$

Linearization (partials at the estimate; $\bar{a}'$ does not depend on the state $\bar{a}_w$, so $\partial y_2 / \partial \bar{a}_w$ is exactly 1)

$$\frac{\partial y_2}{\partial \bar{a}_w} = 1, \quad \frac{\partial y_2}{\partial b_w} = -\nabla_d \bar{w}_d[k] J_{b_w}[k], \quad \frac{\partial y_2}{\partial p_w} = -\nabla_d \bar{w}_d[k] J_{p_w}[k], \quad \frac{\partial y_2}{\partial \bar{w}_s} = -\nabla_d \bar{w}_d[k] J_{w_s}[k]$$

Output equation ($\mathbf{H}$ is the Jacobian; row 2 is nonlinear in $\mathbf{x}$, so $\mathbf{H}\mathbf{x}$ is its linearization, not an identity)

$$\mathbf{y}[k] = \mathbf{H}[k] \mathbf{x}[k] + \mathbf{r}[k], \quad \mathbf{r}[k] = \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}$$

$$\mathbf{H}[k] = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & -\nabla_d \bar{w}_d J_{b_w} & -\nabla_d \bar{w}_d J_{p_w} & -\nabla_d \bar{w}_d J_{w_s} \end{bmatrix}$$

Innovation

$$\begin{aligned} e_{y_1}[k] &= y_1[k] - \delta \hat{w}_1[k] = e_{\delta \bar{w}_1}[k] + n_w[k] \\ e_{y_2}[k] &= y_2[k] - \hat{a}_w[k] + \hat{a}'[k] \nabla_d \bar{w}_d[k] \\ &= e_{\bar{a}_w}[k] - \nabla_d \bar{w}_d[k] [J_{b_w}[k] e_{b_w}[k] + J_{p_w}[k] e_{p_w}[k] + J_{w_s}[k] e_{\bar{w}_s}[k]] + n_a[k] \end{aligned}$$

8 Process and measurement noise

Process noise Q (full $\text{Var}(\varepsilon_w)$ container — thermal history and n_w feedthrough included, only the cross-step correlations of ε_w dropped; revision 2026-08-01, supersedes the current-step-only container; see ref)

$$\begin{aligned} q_3[k] &= -\bar{a}_w[k] \bar{f}_T[k], \quad q_4[k] = \hat{a}'[k] \bar{a}_w[k] \bar{f}_T[k] = -\hat{a}'[k] q_3[k] \\ \gamma(\bar{w}) &= \gamma_N c(\bar{w}), \quad \sigma_{\bar{f}_T}^2 = a_o^2 \frac{4k_B T \gamma(\bar{w})}{\Delta t}, \quad \bar{a}_w = \frac{\Delta t}{a_o \gamma(\bar{w}) R} \implies \text{Var}(\bar{a}_w \bar{f}_T) = \frac{4k_B T a_o}{R} \bar{a}_w \end{aligned}$$

$$Q_{33}[k] = \text{Var}(\varepsilon_w[k]) = \frac{4k_B T a_o}{R} \left(\bar{a}_w[k] + (1 - \lambda_c)^2 \sum_{i=1}^d \bar{a}_w[k-i] \right) + (1 - \lambda_c)^2 \sigma_{n_w}^2$$

$$Q_{34}[k] = Q_{43}[k] = -\hat{a}'[k] Q_{33}[k], \quad Q_{44}[k] = \hat{a}'[k]^2 Q_{33}[k]$$

$$Q_{55} = Q_{66} = Q_{77} = 0 \quad (b_w, p_w, \bar{w}_s \text{ constant})$$

All other entries vanish; the gain block is rank one, since $q_4 = -\hat{a}'[k] q_3$. (run time: entries at the current estimate)

Measurement noise $R = \text{diag}(R_1, R_2)$ ($a_{cov} = \text{EWMA coefficient}$, $IF_{var} = \text{MA}(d)$ inflation factor of $\hat{\sigma}_{\delta\bar{w}_r}^2$; y_2 treated as white here — whitened treatment in the ref notes)

$$R_1 = \sigma_{n_w}^2$$

$$\sigma_{\delta\bar{w}_r}^2[k] = C_{dpmr} \frac{4k_B T a_o}{R} \bar{a}_w[k] + C_n \sigma_{n_w}^2 = C_{dpmr} \frac{4k_B T a_o}{R} (\bar{a}_w[k] + \xi)$$

$$\xi := \frac{C_n}{C_{dpmr}} \frac{\sigma_{n_w}^2}{4k_B T a_o / R}$$

$$\bar{a}_{wm}[k] = \frac{\hat{\sigma}_{\delta\bar{w}_r}^2[k] - C_n \sigma_{n_w}^2}{C_{dpmr} (4k_B T a_o / R)}$$

$$\text{Var}(\hat{\sigma}_{\delta\bar{w}_r}^2) = K_{var} IF_{var} (\sigma_{\delta\bar{w}_r}^2)^2, \quad K_{var} = \frac{2a_{cov}}{2 - a_{cov}}$$

$$R_2[k] = K_{var} IF_{var} (\bar{a}_w[k] + \xi)^2 + d Q_{44}[k]$$

(C_{dpmr} , C_n : `Cdpmr_Cn_derivation.tex` — stationary, noise-only derivation; the former “defect 2” motion excess was RETRACTED 2026-08-01 ($N=48$: +1.33% n.s.); see ref)

MA(2) augmentation (2026-08-01; supersedes the white container above. ε_w is MA(2) across steps — the d -step loop delay makes consecutive ε_w share thermal kicks; the white container matches the per-step variance but understates the zero-frequency power by $(1 + 2(1-\lambda_c))^2 / (1 + 2(1-\lambda_c)^2) = 2.17$ at $\lambda_c = 0.7$; verification and ablation in ref)

$$m_1[k] := \bar{a}_w[k-1] \bar{f}_T[k-1], \quad m_2[k] := \bar{a}_w[k-2] \bar{f}_T[k-2]$$

$$\varepsilon_w[k] = \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c)(m_1[k] + m_2[k]) - (1 - \lambda_c) n_w[k-d]$$

$$m_1[k+1] = \bar{a}_w[k] \bar{f}_T[k], \quad m_2[k+1] = m_1[k]$$

$$\mathbf{x} \rightarrow [\mathbf{x}, m_1, m_2]^\top \quad (9 \text{ states}; m_1, m_2 \text{ at slots } 8, 9)$$

$$\mathbf{F}_e(3, 8) = \mathbf{F}_e(3, 9) = -(1 - \lambda_c), \quad \mathbf{F}_e(4, 8) = \mathbf{F}_e(4, 9) = \hat{a}'[k] (1 - \lambda_c),$$

$$\mathbf{F}_e(8, \cdot) = \mathbf{0}, \quad \mathbf{F}_e(9, 8) = 1 \quad (\text{rows } 1-7 \text{ otherwise unchanged})$$

Per-step noise (white by construction; the off-diagonal couplings Q_{38} , Q_{48} are load-bearing — a diagonal Q gives $C(1) = (1 - \lambda_c)^2 \sigma_T^2$, $C(2) = 0$, wrong)

$$\mathbf{g}_T = [0, 0, -1, \hat{a}'[k], 0, 0, 0, 1, 0]^\top, \quad \mathbf{g}_n = (1 - \lambda_c)[0, 0, -1, \hat{a}'[k], 0, 0, 0, 0, 0]^\top$$

$$\mathbf{Q}[k] = \frac{4k_B T a_o}{R} \bar{a}_w[k] \mathbf{g}_T \mathbf{g}_T^\top + \sigma_{n_w}^2 \mathbf{g}_n \mathbf{g}_n^\top \quad (\text{rank } 2)$$

Autocovariance identity ($\alpha := 1 - \lambda_c$, $\sigma_T^2 := (4k_B T a_o / R) \bar{a}_w$; the augmented model reproduces exactly, the white container only $C(0)$)

$$C(0) = (1 + 2\alpha^2) \sigma_T^2 + \alpha^2 \sigma_{n_w}^2, \quad C(1) = \alpha(1 + \alpha) \sigma_T^2, \quad C(2) = \alpha \sigma_T^2, \quad C(j \geq 3) = 0$$

$$S(0) = (1 + 2\alpha)^2 \sigma_T^2 + \alpha^2 \sigma_{n_w}^2$$

Initialization (m -block from the same DARE pattern as the position block, now 5×5 over $[\delta \bar{w}_1 \ \delta \bar{w}_2 \ \delta \bar{w}_3 \ m_1 \ m_2]$; $P_{m_1 m_1}[0] = P_{m_2 m_2}[0]$ carried by the DARE steady state at the seed gain)

y_2 self-echo correction (2026-08-01. The variance readout measures the ACTUAL loop, which runs on the applied gain $\hat{a}_{ctrl}$; a control-gain error shifts the loop pole and the residual variance follows it with sensitivity S — the innovation therefore carries only $(1-S)$ of the true-gain deviation, and the uncorrected filter overcounts y_2 information by $1/(1-S)^2$ and partially confirms its own error; verification in ref)

$$\sigma_{\delta \bar{w}_r}^2 = \sigma^2(\bar{a}_w, \hat{a}_{ctrl}), \quad S := \frac{\partial \ln \sigma_{\delta \bar{w}_r}^2}{\partial \ln \hat{a}_{ctrl}}$$

$$E[y_2[k]] = (1 - S) \bar{a}_w[k - d] + S \hat{a}_{ctrl}[k] + \dots$$

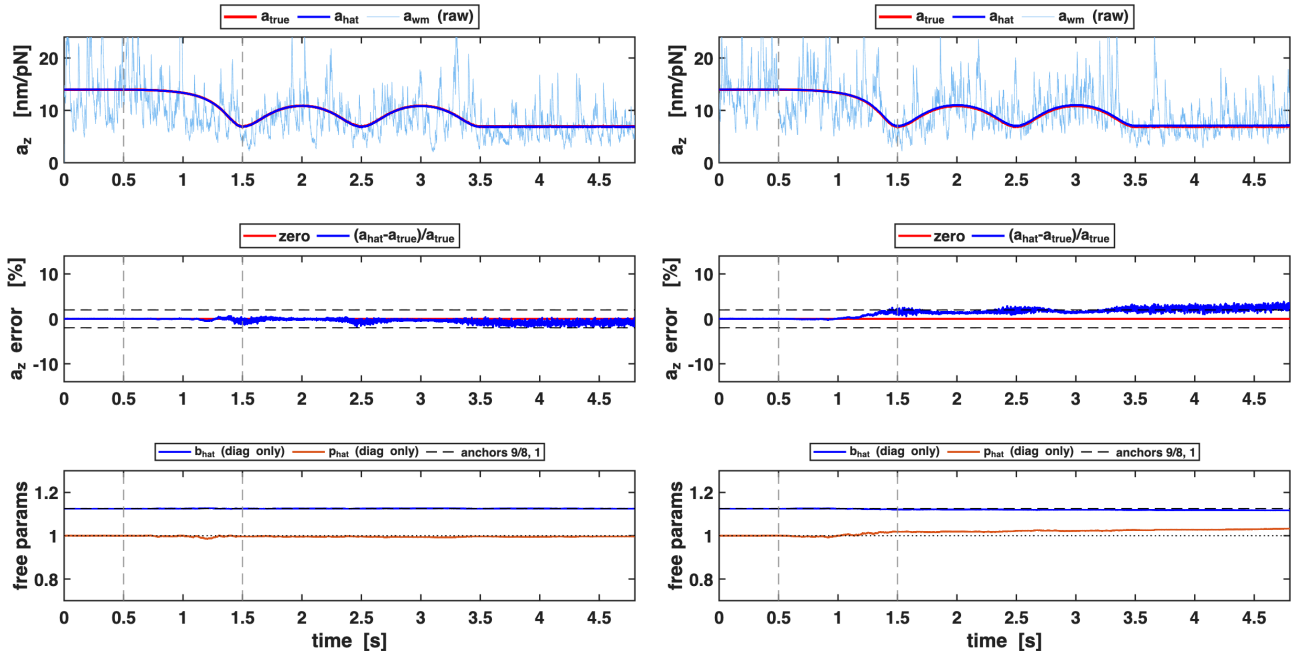
$$\implies \mathbf{H}_{2,:} \rightarrow (1 - S) \mathbf{H}_{2,:}, \quad y_{2,pred} = \hat{a}_w - (1 - S) \hat{a}' \nabla_d \bar{w}_d$$

$$S = \frac{S_T \bar{a}_w + S_n \xi}{\bar{a}_w + \xi}, \quad S_T, S_n \text{ from the exact 6-state Lyapunov covariance of the mismatched loop}$$

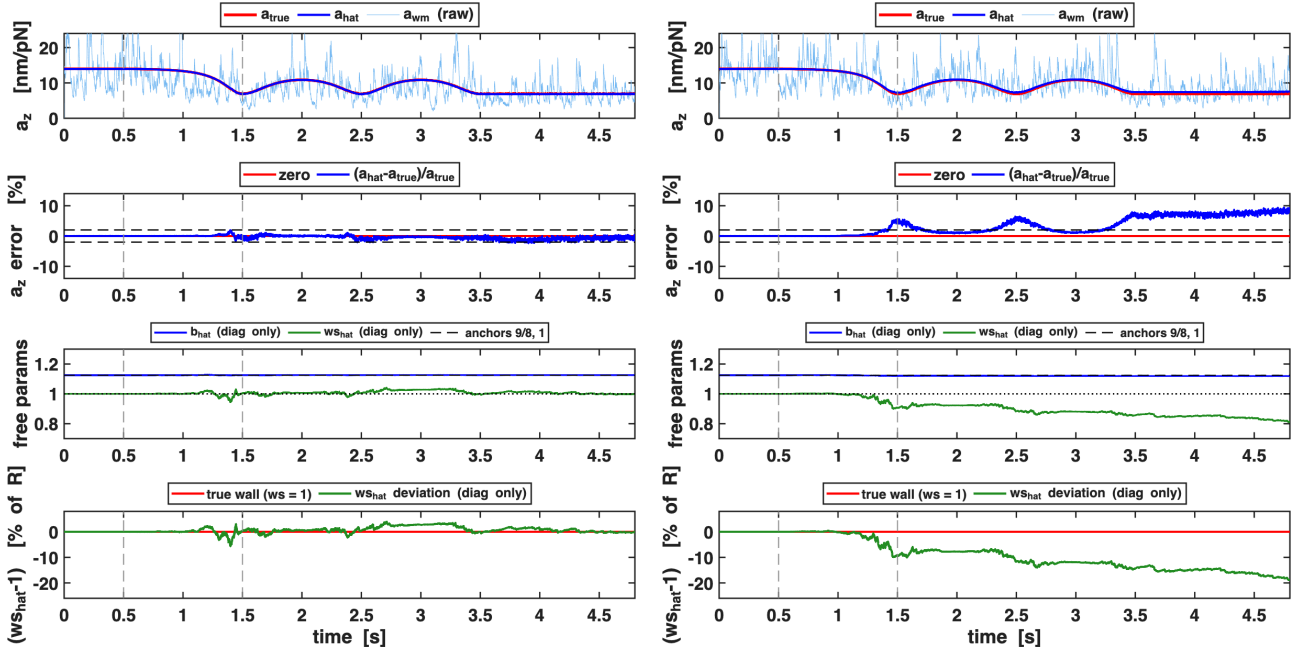
(S_T far field 0.319; paired-forcing measurement 0.323 ± 0.043 ; $S(\bar{w})$ near-constant 0.310–0.319 on the envelope; R_2 unchanged — the noise floor is on the actual variance level)

Simulation figures

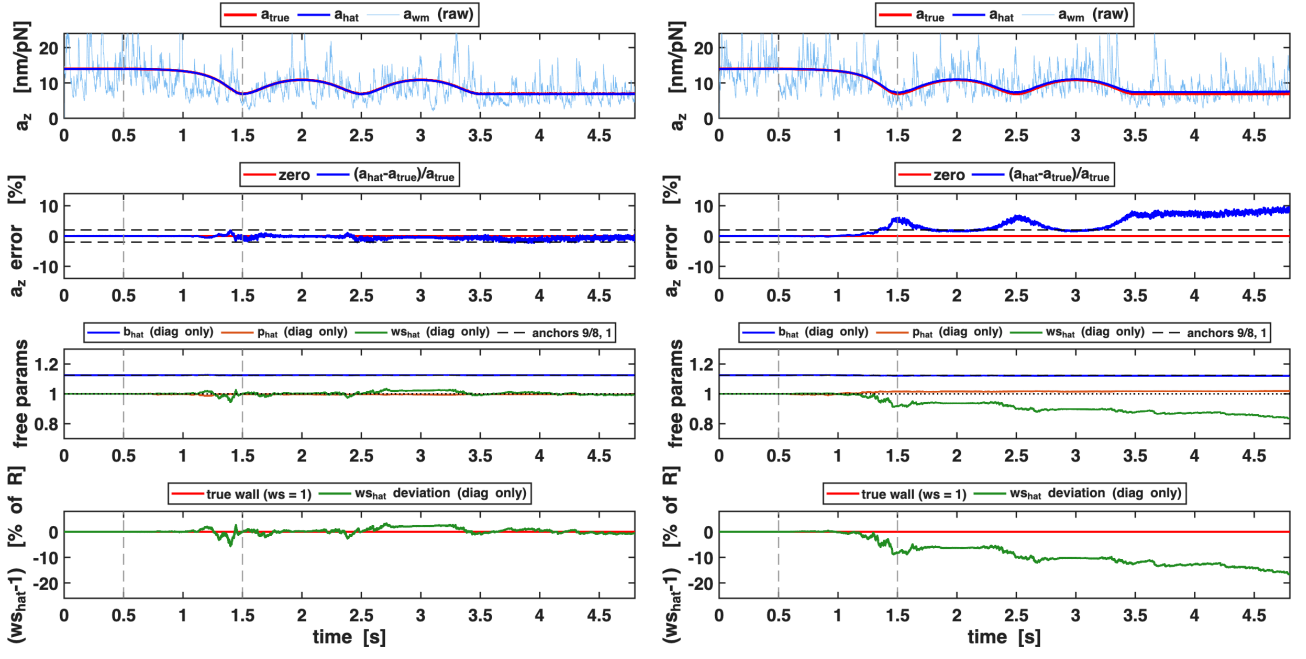
(canonical scenario, z axis; per panel: gain tracking, relative gain error, free parameters)

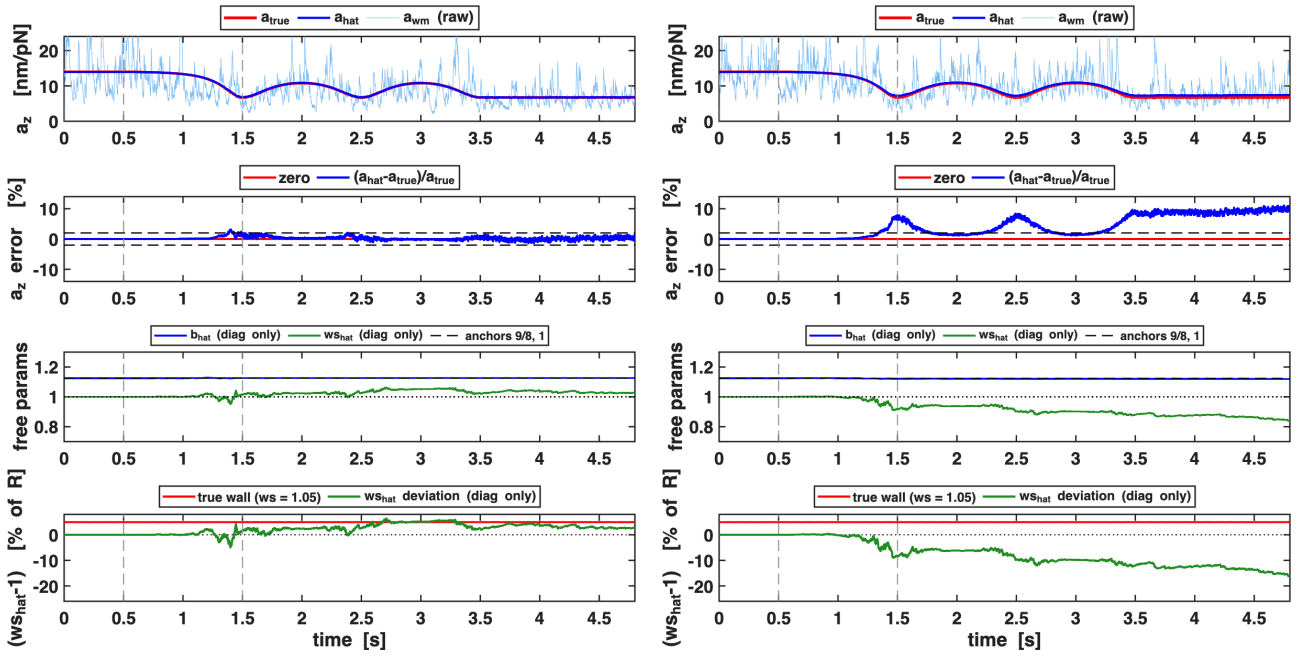


Config 1 — estimate (b, p) , w_s locked at 1: seed 7 (left) vs seed 11 (right).

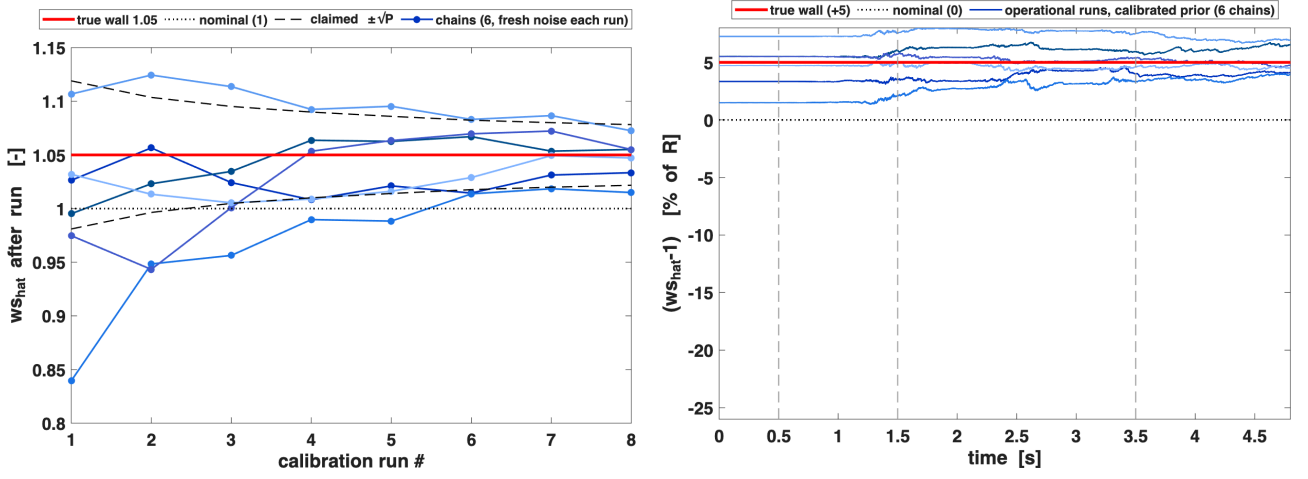


Config 2 — p locked at 1, estimate (b, w_s) : seed 7 vs seed 11.

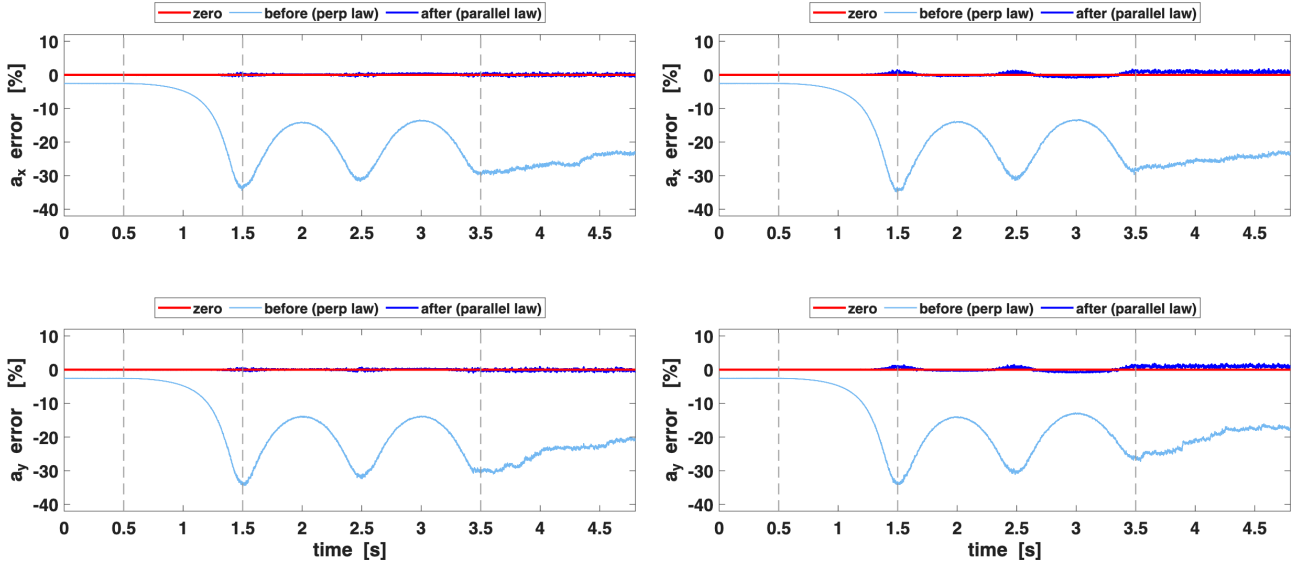
Config 3 — estimate (b, p, w_s) : seed 7 vs seed 11.



Config 4 — config 2 with the TRUE wall injected at $\bar{w}_s = 1.05$ (plant-side shift, S11 protocol): seed 7 vs seed 11.



Config 5 — $\bar{w}_s$ as a calibration constant: sequential posterior handoff (run n 's $(\hat{w}_s, \sqrt{P_{w_s}}) =$ run $n+1$'s prior), 6 chains $\times$ 8 runs, fresh noise each run, true wall 1.05 (left); operational run on the calibrated prior — cross-chain $\hat{w}_s$ error std 1.3%, descent peak 1.69% ($\approx$ the config-1 locked floor) (right).



Config 6 — Stage 1a, x/y gain estimation before (perpendicular law, $b=9/8$) vs after (derived parallel law, S1 parallel block): 6-seed means desc $34.8/34.1 \rightarrow 0.92/0.86\%$, hold $-25.0/-22.6 \rightarrow -0.04/+0.01\%$; z bit-identical. Seed 7 (left) vs seed 11 (right).